

```
from diffusers import StableDiffusionPipeline
import torch
from PIL import Image

# Load the Stable Diffusion model from Hugging Face
model_id = 'CompVis/stable-diffusion-v1-4'
pipeline = StableDiffusionPipeline.from_pretrained(model_id)
pipeline = pipeline.to('cuda') if torch.cuda.is_available() else pipeline

# Function to generate and save images from text prompts
def generate_and_save_image(prompt: str, image_name: str):
    print(f'Generating image for: {prompt}')
    image = pipeline(prompt, height=resolution, width=resolution)
    image.save(f'{image_name}.png')
    image.show()
    print(f'Image saved as {image_name}.png')

# Generate image for the initial prompt
generate_and_save_image('A serene mountain landscape with snow-capped peaks and a calm lake')

# Experiment with additional prompts
prompts = [
    'A cute cat wearing a top hat',
]

for i, prompt in enumerate(prompts, start=1):
    generate_and_save_image(prompt, f'prompt_image_{i}.png')

print('Image generation completed.')
```

mountain_landscape.png

pi ...





The cache for model files in Transformers v4.22.0

0/0 [00:00<?, ?it/s]

```
/usr/local/lib/python3.11/dist-packages/huggingface
The secret `HF_TOKEN` does not exist in your Colab
To authenticate with the Hugging Face Hub, create
You will be able to reuse this secret in all of yo
Please note that authentication is recommended but
warnings.warn(
```

model_index.json: 100% 541/541 [00:00<00:00, 35.4kB/s]

Fetching 16 files: 100% 16/16 [00:29<00:00, 1.94s/it]

model.safetensors: 100% 1.22G/1.22G [00:16<00:00, 26.8MB/s]



text_encoder%2Fconfig.json: 100% 592/592 [00:00<00:00, 16.4kB/s]



(...)oints%2Fscheduler_config.json: 100% 209/209 [00:00<00:00, 2.75kB/s]

checkpoint.json: 100%



scheduler%2Fscheduler_config.json: 100% 313/313 [00:00<00:00, 3.13kB/s]



(...)ure_extractor%2Fpreprocessor_config.json: 100% 342/342 [00:00<00:00, 3.42kB/s]



model.safetensors: 100% 492M/492M [00:09<00:00, 82.5MB/s]



safety_checker%2Fconfig.json: 100% 4.56k/4.56k [00:00<00:00, 4.56kB/s]



tokenizer%2Fmerges.txt: 100% 525k/525k [00:00<00:00, 2.15kB/s]



tokenizer%2Ftokenizer_config.json: 100% 806/806 [00:00<00:00, 8.06kB/s]



tokenizer%2Fspecial_tokens_map.json: 100% 472/472 [00:00<00:00, 4.72kB/s]



tokenizer%2Fvocab.json: 100% 1.06M/1.06M [00:00<00:00, 3.15MB/s]



diffusion_pytorch_model.safetensors: 100% 3.44G/3.44G [00:29<00:00, 1.15GB/s]



unet%2Fconfig.json: 100% 743/743 [00:00<00:00, 6.26kB/s]

